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Interfacial insight from *Operando* XAS/TEM for magnesium metal deposition with borohydride electrolytes.

Timothy S. Arthur^{*†}, Per-Anders Glans[‡], Nikhilendra Singh[†], Oscar Tutusaus[†], Kaiqi Nie[‡], Yi-Sheng Liu[‡], Fuminori Mizuno[†], Jinghua Guo^{‡‡}, Daan Hein Alsem[©], Norman J. Salmon[©] and Rana Mohtadi[†]

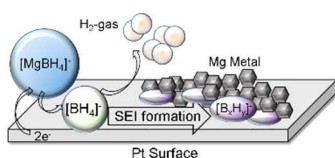
[†]Toyota Research Institute of North America, 1555 Woodridge Ave, Ann Arbor, MI 48103

[‡]Advanced Light Source, Lawrence Berkeley National Laboratory, One Cyclotron Rd, Berkeley, CA 94720

^{‡‡}Department of Chemistry and Biochemistry, University of California, Santa Cruz, CA 95064

[©] Hummingbird Scientific, 2610 Willamette Dr NE, Lacey, WA 98516

TOC



Summary

For the first time, a solid electrolyte interphase (SEI) is observed for magnesium metal deposition from borohydride electrolytes by using *operando* X-ray absorption and transmission electron microscopy. The key mechanistic steps are boron-hydrogen cluster formation and hydrogen gas evolution. Future Mg anode work can tailor interfaces to access new classes of magnesium electrolytes.

ABSTRACT

Operando soft X-ray absorption spectroscopy (sXAS) and transmission electron microscopy (TEM) of the electrochemical deposition of magnesium metal from borohydride electrolytes is presented. The analysis identifies the active component for the deposition of magnesium metal as the contact-ion pair $[\text{Mg}(\mu\text{-H})_2\text{BH}_2]^+$. Additionally, the formation of boron clusters accompanied with H_2 -gas formation accompany the deposition of Mg metal, and play a vital role in the formation of a solid-electrolyte interphase (SEI) on the surface. To use Mg metal as a viable anode for future battery systems, understanding the interfacial components and chemistry is key.

Introduction

The quest for the next energy storage system has encouraged exploratory research into various electrochemically active materials beyond Li-ion.¹ Li-Oxygen, Li-Sulfur, anion and multivalent systems, such as Mg^{2+} , Al^{3+} and Ca^{2+} , have dominated recent research strategies, although all systems face considerable challenges.^{2,3} To realize Mg batteries, cathodes which undergo the reversible reduction and oxidation reactions (discharge and charge, respectively) remain a demanding goal.^{4,5} However, the benefits of a magnesium (Mg) battery over lithium (Li) are based on the metal anode utilization, where the more positive reduction potential, -3.05 V (Li) vs -2.37 V (Mg) vs SHE, is balanced with the divalent nature of the cation, $1\text{e}^- / \text{Li}^+$ vs $2\text{e}^- / \text{Mg}^{2+}$, and the high abundance of Mg-based compounds found in the earth's crust.⁶ Additionally, Mg-metal has been shown to deposit non-dendritically, a major challenge for Li-metal realization, however, this assertion must be understood in the context of the anode and electrolyte interface and interphase. Aurbach *et al.* have developed high-performing magnesium electrolytes based on magnesium organochloroaluminates or organoborates dissolved in tetrahydrofuran, which do not form a surface passivating film and allow for efficient

Mg deposition / stripping.⁷⁻⁹ Recent results have countered the requirement to maintain a “bare” Mg-metal surface as interest in the bis(trifluoromethylsulfonylimide) salt, Mg(TFSI)₂, dissolved in glymes has gained momentum and shed light on stability of electrolytes in contact with Mg-metal. In addition, Mohtadi *et al.* developed electrolytes based on a mixture of borohydride salts, Mg(BH₄)₂ and LiBH₄, and have recently extended the oxidative stability of the electrolyte by coupling *mono*-anionic, *closo*-carborane clusters [CB₁₁H₁₂][−] with the Mg²⁺.¹⁰⁻¹³ The growth of the magnesium electrolyte family necessitates continued investigation of the anode / electrolyte interface to determine if magnesium deposition / dissolution can occur in the presence of surface films.

Magnesium borohydride provides a segue between classic electrolytes based on Grignard reagents and the new electrolytes based on simple magnesium salts. Much like organohaloaluminates and organoborates, borohydrides are reactive towards moisture which eliminates the influence of water (H₂O) on solvation/desolvation of Mg²⁺ and surface species that hinder the deposition and dissolution of Mg-metal. Borohydrides also provide a route to forming active, non-halogenated electrolytes, as the Cl[−] anion has been shown to corrode cell components around ~2.0 V vs Mg.¹⁴ Even though the low oxidation potential of Mg(BH₄)₂, 1.7 V vs Mg, is a well-documented hindrance to final high voltage applications, few studies have captured the important facets of the anode / electrolyte interface.¹⁵ Progress in cell fabrication to utilize soft X-ray absorption spectroscopy (sXAS) have been pivotal to introduce *operando* analysis of representative electrochemical devices.¹⁶ Our previous work illustrated the functionality of *operando* cells to investigate the deposition and dissolution of Mg-metal from the recrystallized Mg-dimer [Mg₂(μ-Cl)₃·(THF)₆](AlCl₄), which supported the hypothesis of the active [Mg-Cl]⁺ cation as the active, reducible species at the anode / electrolyte interface.^{17,18} Importantly, the study also confirmed the lack of an appreciable solid-electrolyte interphase (SEI) on the anode surface after deposition and dissolution. SEI formation on the surface of

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2
3 anodes, such as Li-metal, plays a major role in morphology, rate and cycle-life of lithium
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5 batteries, yet is vital to stabilize the resistance at the anode / electrolyte interphase. In contrast,
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7 we have observed a growing interfacial impedance at the Mg-metal / electrolyte interface as the
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9 cell is held at open-circuit potential, which directly lowers the energy density as the cell moves
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11 from the electrochemically static to a dynamic state.¹⁹ Considering practical function of a
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13 battery involves persistent switching between the “on” and “off” state, rather than the constant-
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15 current lab-scale examination; thus, interfacial and interphase understanding is tremendously
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17 important for Mg metal anodes.
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21 Here, we combine *operando* electrochemical-synchrotron soft X-ray absorption (sXAS) and -
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23 transmission electron microscopy (TEM) with electrochemical studies of Mg-metal in
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25 magnesium borohydride-based electrolytes to gain a fundamental understanding of the
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27 interphase which permits the deposition and dissolution of Mg-metal. For the first time, the
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29 observation of an SEI layer is observed in a magnesium borohydride electrolyte during metal
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31 deposition and dissolution.
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Experimental

Mg(BH₄) electrolytes:

Magnesium borohydride (Mg(BH₄)₂, 95%), lithium borohydride (LiBH₄, 90%), and sodium borohydride (NaBH₄, 95%), anhydrous tetrahydrofuran (THF), and dimethoxyethane (DME) were purchased from Sigma–Aldrich. The concentrations used for each electrolyte is as follows:

- 1) Mg borohydride THF: 0.5 M Mg(BH₄) / THF
- 2) Mg borohydride DME: 0.1 M Mg(BH₄) / DME
- 3) Mg:Li 1:1 borohydride: 0.2 M Mg(BH₄) / DME + 0.2 M LiBH₄ / DME
- 4) Mg:Li 1:3 borohydride: 0.2 M Mg(BH₄) / DME + 0.6 M LiBH₄ / DME
- 5) Mg:Na 1:3 borohydride: 0.2 M Mg(BH₄) / DME + 0.6 M NaBH₄ / DME

Synthesis of (Hep₄N)₂(B₁₂H₁₂) [1]:

A solution of K₂B₁₂H₁₂ (4.462 g, 20.3 mmol) in H₂O (10 ml) was added to a stirring solution of Hep₄NBr (19.41 g, 39.6 mmol) in acetone/H₂O (9:8, 170 ml). An oily phase separated immediately. The mixture was stirred for 30 mins and solvent volume was reduced to approximately 75 ml by rotary evaporation. The white solid is collected by filtration, washed with a large excess of H₂O, followed by a large excess of hexane. It was then dried under vacuum at 50 °C until constant weight to yield **1** as a white solid (18.9 g, 99%). m.p. 71–72 °C; ¹H NMR (δ, DMSO-*d*₆): 3.15 [m, 16H, N-CH₂], 1.56 [m, 16H, CH₂], 1.46–0.46 (br, 12H, BH), 1.35–1.22 [m, 64H, CH₂], 0.87 [t, 24H, CH₃]. ¹¹B NMR (δ, DMSO-*d*₆): 15.5 [d, BH, J(BH)=120].

Synthesis of Mg(G₂)₂(B₁₂H₁₂) [G₂ = diglyme]:

Under an atmosphere of argon, a solution of MgTfSI₂ (586.0 mg, 1.0 mmol) in diglyme (4 ml) was added to a stirring solution of **1** (966.7 mg, 1.0 mmol) in diglyme (10 ml). The white suspension thus obtained was stirred at room temperature for 1 h. The white solid was collected by filtration, washed with diglyme (2×4 ml), then Et₂O (2×10 ml), and dried under vacuum to obtain 429 mg (99%) of Mg(G₂)₂(B₁₂H₁₂) as a white solid. ¹H NMR (δ, DMSO-*d*₆, 60 °C): 3.53 [m, 8H, CH₂], 3.44 [m, 8H, CH₂], 3.26 [s, 12H, CH₃], 1.24–0.51 [br, 12H, BH]. ¹¹B NMR (δ, DMSO-*d*₆, 60 °C): 15.5 [d, BH, J(BH)=123].

Operando electrochemical/sXAS:

Figure S1a is a cross-sectional drawing of the *operando* electrochemical/sXAS cell, which builds upon previous liquid-compatible cell.¹⁷ The material of interest was sealed with a Teflon O-ring (v) behind a 100 nm thick Si₃N₄ window (viii) etched into a 1 cm x 1 cm silicon wafer (vii). The Si₃N₄ window was sputtered with 40 nm of platinum on one side to serve as a working electrode (vi). The geometry of the cell is ideal to probe the electrochemical interface through the thin platinum layer (Figure S1b). Two additional electrodes, Mg foil and a platinum wire (ii and iii), were inserted into the base of the cell.

Operando electrochemical/TEM:

All TEM analysis was carried out on a JEOL JEM-2100 LaB6 Microscope, operating at 200 kV with a Direct Electron DE-12 3k x 4k camera. All experiments presented in this manuscript were accomplished using a Hummingbird Scientific Liquid-Electrochemistry TEM Holder, outfitted with platinum electrode Electrochemistry Chips (50 nm thick SiN_x) and 250 nm Spacer Chips (50 nm thick SiN_x). Additionally, chemical resistant O-rings (inner and outer) were utilized in order to negate the effect of the electrolyte on the O-ring seals. All handling and storage of the electrolyte was carried out within an Argon-filled glove box. The TEM Holder itself was first assembled outside the glove box (inclusive of all tip components – chips and O-rings). Once assembled the Holder was allowed to dry at 60 °C for 1 hour, prior to its introduction to the glove box. Once within the glove box, the electrolyte was introduced into the Holder and an additional aliquot was kept in a gas-tight syringe, attached to the Holder electrolyte tubing. Once removed from the glove box, the Holder was leak-tested to ensure that the electrolyte was safely contained within the Holder and not leaking. Upon successfully passing the leak-test, the Holder was introduced to the TEM mentioned above, and the syringe attached to a standard syringe pump to allow for a continuous supply of electrolyte to the Holder tip and Electrochemistry Chip, at a flow rate of 2 μL/min. Cyclic Voltammograms (CV) were recorded at 2 mV s⁻¹, and the data reported within this manuscript was captured at the end of the reductive sweep of the CV at room temperature.

sXAS Data and Analysis:

The Mg K-edge was detected by fluorescence on the beamline 6.3.1.2 ISAAC end station at Lawrence Berkeley National Lab. At the Mg absorption edge (45° incidence geometry), the 100 nm SiN_x window was controlled to maintain a transmission of 94% of X-ray, 40 nm thick of Pt layer transmitted 78%, which amounted to ~72% transmission for incoming X-ray to the liquid. Scattered X-rays from liquid travel through the same layers with ~72% transmission to the diode detector. All energies were calibrated with the MgO standard. XAS data was processed with the Athena program from the HORAE software package. Background removal and normalization of the energies to the edge step were performed for each spectrum. For Fourier transform of the EXAFS region, the range for the k³-weighted spectra was 2.0–6.0 Å⁻¹. The B K-edge was measured at BL 8.0.1 Wet-RIXS at the Advanced Light Source. Samples were prepared on platinum foil and loaded into the analysis chamber without exposure to air. All Energies were calibrated to a BN standard.

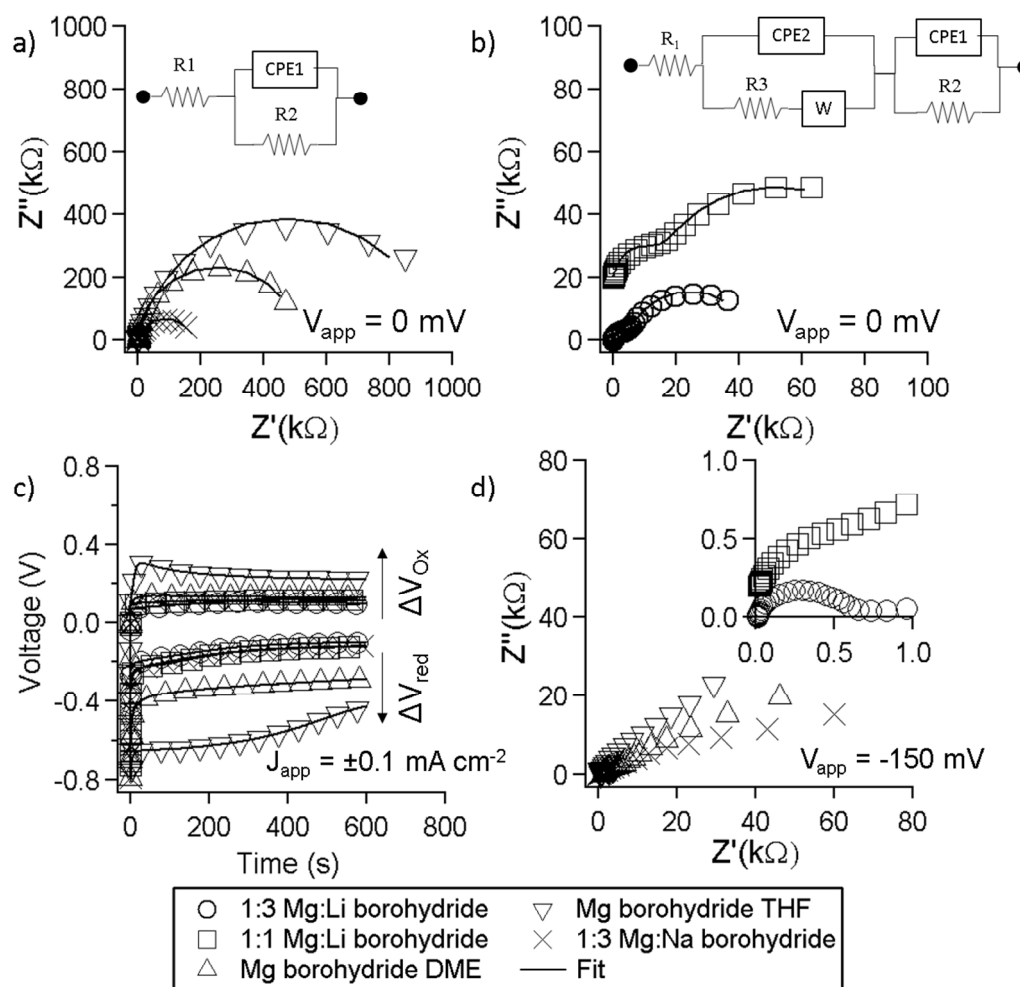
Results and Discussion

Figure 1. a), b) EIS spectra and c) Galvanic discharge and charge of Mg-Mg symmetric cells. d) EIS spectra of Mg-Mg cells under a -150 mV bias.

Electrochemical impedance spectroscopy (EIS) is an excellent tool to ascertain the difference in electrochemistry between variations of magnesium electrolytes. When fabricated in a symmetric geometry, Mg metal as both the anode and cathode, such analysis can isolate electronic, interfacial and bulk resistances specifically for the anode / electrolyte interface. Figure 1a) and b) are the EIS Nyquist response of symmetrical cells containing $\text{Mg}(\text{BH}_4)_2$ ($V_{\text{app}} =$ applied voltage bias, 0 mV), and the curve-fitting analysis based on the corresponding electrical models (*inset*). Quantitative values for the circuit elements are tabulated in the **Supporting Information – Table 1**. As extensively studied by Shao *et al.*²⁰ and Mohtadi *et al.*²¹, solvation of Mg^{2+} significantly impacts the solubility limit of $\text{Mg}(\text{BH}_4)_2$ in ethereal solvents, 0.1 M in DME, as well as the electrochemical properties of the electrolytes due to changes in the solution equilibrium between $\text{Mg}[(\mu\text{-H})_2\text{BH}_2]_2$, $\text{Mg}[(\mu\text{-H})_2\text{BH}_2]^+$ and Mg^{2+} . Therefore, the focus of the research is to probe the interfacial aspects of the deposition while the concentration of the electrolytes of magnesium borohydrides is maintained from the initial work of Mohtadi *et al.*¹⁰. As expected, bulk resistance of the electrolytes (R_1) decreases as the amount of dissolved salt increases in the electrolyte, which may be mimicked by the addition of a supporting salt to the electrolyte solution. The interfacial resistance (R_2) is defined as the low-frequency intercept in the Nyquist response. Indeed, dissolving 3 molar equivalents of sodium borohydride (0.6 M NaBH_4) further reduces the bulk resistance of the electrolyte $R_1 = 23.6 \, \Omega$ and interface resistance $R_2 = 174 \, \text{k}\Omega$. Dissolution in 1,2-dimethoxy ethane (0.1 M $\text{Mg}(\text{BH}_4)_2$ / DME), as opposed to tetrahydrofuran (0.5 M $\text{Mg}(\text{BH}_4)_2$ / THF), serves to decrease the interfacial resistance of the electrolyte from R_2 (THF) = 950 k Ω to R_2 (DME) = 503 k Ω . The change in interface resistance is likely due to the increased presence of an active species, $\text{Mg}[(\mu\text{-H})_2\text{BH}_2]^+$ and / or Mg^{2+} , close to the electrochemical interface.

Interestingly, lithium borohydride (LiBH_4) addition alters the interfacial impedic Nyquist response by the introduction of an adsorbed species with surface diffusivity. Hwang *et al.*²²

evidenced the high surface mobility and the specific capability of LiBH_4 to form solid-solutions with alkali or alkali-earth borohydrides, which is captured in the resistive element R3. Therefore, LiBH_4 not only reduces the bulk resistance of the electrolyte, R1 (1:3 Mg:Li) = 8.75 Ω and interface resistance R3 (1:3 Mg:Li) = 50 k Ω , but also enhances the surface mobility of adsorbed species at the interface R2 (1:1 Mg:Li) = 12.5 k Ω and R2 (1:3 Mg:Li) = 4.1 k Ω .

The variation in interfacial species and resistances directly influences the galvanic oxidation and reduction, Mg metal stripping and plating, respectively. As shown in Figure 1c), the overpotential (ΔV_{red}) for reduction is highest for 0.5 M $\text{Mg}(\text{BH}_4)_2$ / THF and lowest for 1:3 Mg:Li / DME, however, the ΔV_{ox} for the first oxidation is much lower for every electrolyte. The asymmetry in the energy required for stripping and plating magnesium is indicative of an electrochemically induced interface formation during the initial galvanic step, commonly known as SEI formation for Li-ion systems. During a dynamic EIS analysis, $V_{\text{app}} = -150$ mV, there is a noticeable decrease in resistance for all electrolytes highlighting an electrochemical activation of the magnesium surface (*vide supra*). The intriguing properties of the electrochemical interface warrants an elucidation of the chemical constituents at the anode / electrolyte interface.

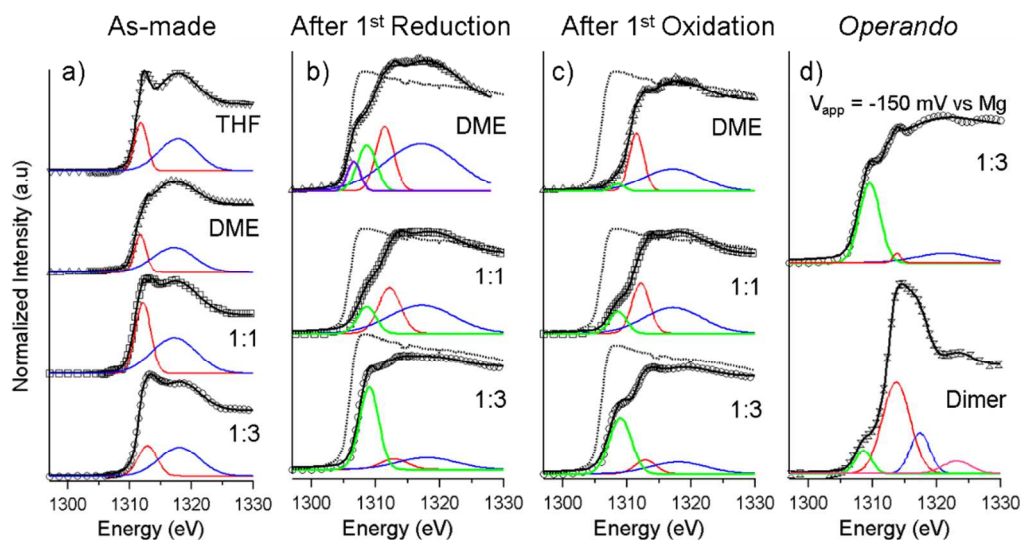


Figure 2. Mg K-edge XANES of borohydride electrolytes a) As-made, b) after the 1st-reduction and c) after the 1st oxidation. d) Mg K-edge spectroscopy of the cell under *operando* $V_{\text{app}} = -150$ mV Mg applied

to the interface. The solid black line is the fit obtained from the addition of the red, blue, green and purple peaks. The dashed black line is a Mg-metal foil reference.

Investigations into the interfacial components for battery anodes has extensively increased over the past decade, resulting in significant improvements to the capacity, cycle life, and rate for energy storage. sXAS is an element specific technique sensitive to changes in structure and oxidation state of the target molecule. Recent advances in cell fabrication has enabled *operando* sXAS analysis of the electrochemical interface during magnesium metal deposition and stripping (**Figure S1**). Figure 2 shows the Mg K-edge X-ray Absorption Near Edge Structure (XANES) of magnesium borohydride-based electrolytes after, or during, electrification of the Pt/electrolyte interface. As shown in Figure 2a), $\text{Mg}(\text{BH}_4)_2$ dissolved in THF, DME and with the addition of LiBH_4 is composed of two peak structure at 1313 eV and 1317 eV, indicating the similar bulk constituents of the electrolytes but in varying concentrations, as proposed by previous analysis of the bulk properties of these electrolytes. For the purpose of the interfacial analysis highlighted in this work, there is a significant change to the Mg K-edge XANES upon the 1st electrochemical reduction of the interface (Figure 2b). The appearance of two additional peaks signal the deposition of multiple magnesium species onto the electrochemical interface. The presence of Mg-metal is shown by the shift of the Mg K-edge to 1303 eV, and referenced by a Mg-metal foil (dashed black line) and a new intermediate peak at 1307 eV (henceforth Mg_{SEI}) is present in addition to the characteristic electrolyte signals. With the addition of LiBH_4 to the electrolyte, the Mg_{SEI} peak centered at 1307 eV becomes more prominent at the interface, while the signal for magnesium metal is diminished. The loss of the magnesium metal signal at 1303 eV cannot be interpreted as a lack of magnesium metal during reduction, rather, due to the geometry of the *operando* cell, the uniformity and thickness of the Mg_{SEI} at 1307 eV may inhibit the incident X-rays or detection of the fluorescent signal. Indeed, previous researchers have clearly shown the ability of the $\text{Mg}(\text{BH}_4)_2/\text{LiBH}_4$ mixtures to deposit magnesium metal onto

the surface of platinum^{10,20}. Coupling with the EIS results indicate that the observed fast surface diffusion afforded by LiBH₄ plays a large role in the uniformity of thickness of the intermediate magnesium species. Figure 2c) reveals that the deposition of Mg_{SEI} is electrochemically irreversible, likely due to an additional chemical decomposition of the solvent or the BH₄⁻ anion (*vide infra*). However, as shown by the lack of magnesium metal peak at 1303 eV in the electrochemically oxidized Mg(BH₄)₂ / DME electrolyte, the Mg_{SEI} species likely displays Mg²⁺-ion conductivity to allow for the reversible Mg metal deposition/stripping. Therefore, the key to decoding the electrochemical properties of magnesium borohydrides is both the active magnesium species and the chemical constituents of the activated interface.

In our previous *operando* analysis of magnesium deposition from recrystallized [Mg₂(μ-Cl)₃·(THF)₆](N(Si(CH₃)₃)₂AlCl₃), the presence of an active [Mg-Cl]⁺ species at the interface was necessary for magnesium deposition.^{15,16} As shown in Figure 2d), the position Mg K-edge XANES on-set energy is consistent with the 1307 eV seen for the Mg_{SEI} species. A shift to lower energy for XANES is typically associated with a reduction of the target element (as in Mg²⁺ to Mg⁰), however, the shift may also be interpreted as breaking of the symmetry of the octahedral magnesium environment, permitting the previously parity-forbidden σ → σ* intraionic transition. Therefore, the electronic structure similarities of Mg_{SEI} and the [Mg-Cl]⁺ is convincing evidence that the active component, Mg_{SEI}, is a desolvated-[Mg(μ-H)₂BH₂]⁺ contact-ion pair, which was previously shown by Mohtadi *et al.*¹⁰ to exist in the electrolyte equilibrium. The key difference between the organohaloaluminates and borohydride system is the retention of the active magnesium species at the interface in the latter system, while the [Mg-Cl]⁺ exists transiently at the electrified interface.

Interestingly, deposition of interphase layers at the magnesium anode interface is commonly believed to block, or severely inhibit, magnesium deposition and stripping. Therefore, elucidation of the possible reasons for electrochemical activity from this particular interphase will

guide future design of active magnesium electrolytes. Figure 3a) shows the Extended X-ray Absorption Fine Structure (EXAFS) of the Mg K-edge for an as-prepared $\text{Mg}(\text{BH}_4)_2 : 3 \text{LiBH}_4 / \text{DME}$ electrolyte and after one reduction/oxidation cycle. The first peak in the k^3 -weighted spectra can be assigned to the Mg-H bond for the $[\text{Mg}(\mu\text{-H})_2\text{BH}_2]^+$ contact-ion pair present in the interphase layer. Interestingly, the intensity of the peak representing Mg-O bond from the solvent coordination to the Mg^{2+} cation drastically decreases after electrochemically cycling the interface. The decrease in intensity evidences a loss in solvent coordination at the interphase, and correlates well to the breaking of octahedral symmetry of the magnesium cation. This result implies that solvent coordination to the magnesium cation plays a large role in the electrochemical activity of the electrolyte, rather than simply the concentration of LiBH_4 in the electrolyte. Additionally, Persson *et al.*²³ have shown the lengthening of Mg-O bonds of solvated $[\text{Mg}(\mu\text{-H})_2\text{BH}_2]^+$ with the increased chain-length of glymes, and Buttry *et al.*²⁴ have successfully designed ionic liquids with ether appendages for successful deposition and stripping. Investigating the effects of solvents in the bulk and for interphase formation beyond DME is an interesting avenue of future research.

The morphology of metal anode deposition in the presence of an interphase layer must be considered in respect to dendrite formation of the metal, an internal shorting and safety concern. To investigate the morphology of the magnesium metal deposit, we used *operando* liquid-cell TEM as shown in Figure 3b). In this experiment, we observed the deposition of two separate phases on the surface of the Pt metal electrode. Representative cyclic voltammograms for magnesium deposition and dissolution are shown as **Figure S2**. Due to the heavier mass of magnesium (M.W. = 24.3 g/mol) compared to the proposed SEI components boron (10.8 g/mol), carbon (12.0 g/mol) and oxygen (16.0 g/mol), and H_2 -gas (2.0 g/mol), we expect the magnesium layer to display a darker contrast in the TEM image. Differences in contrast has previous been used by Harris *et al.* to identify different components in the deposition of Li

metal.²⁵ Additionally, the image is unchanged before applying a potential to the working electrodes, indicating the lack-of beam-damage effects to the electrolyte/cell components. The geometry of cell allows for the incident electron beam to perpendicularly penetrate through the SiN_x windows containing the electrolyte for the edge-on observation of the working electrode, which is labeled as “WE” in the image. The TEM images, taken from reduction in a 1:3 Mg:Li electrolyte, visualize the formation of the SEI layer and H_2 -gas beneath the magnesium metal, in accordance with our Mg K-edge spectroscopy. Indeed, gas formation was further evidenced by the pressure increase in the *operando* cells, which resulted in repeated fracture of the thin SiN_x electrochemical chip window during the reductive sweep of the cyclic voltammogram under otherwise constant conditions. *Operando* electron microscopy is a powerful tool to deeply understand the deposition and dissolution of metallic anodes for electrochemical devices.

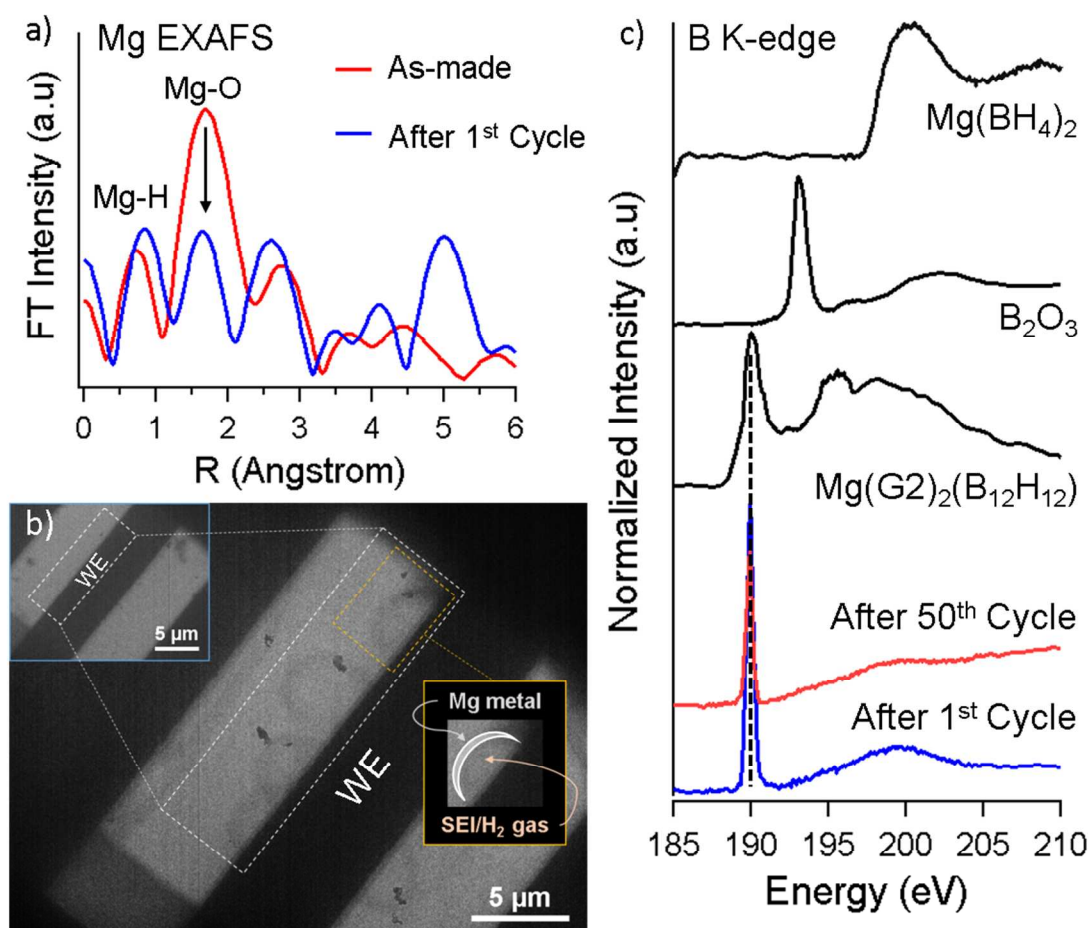
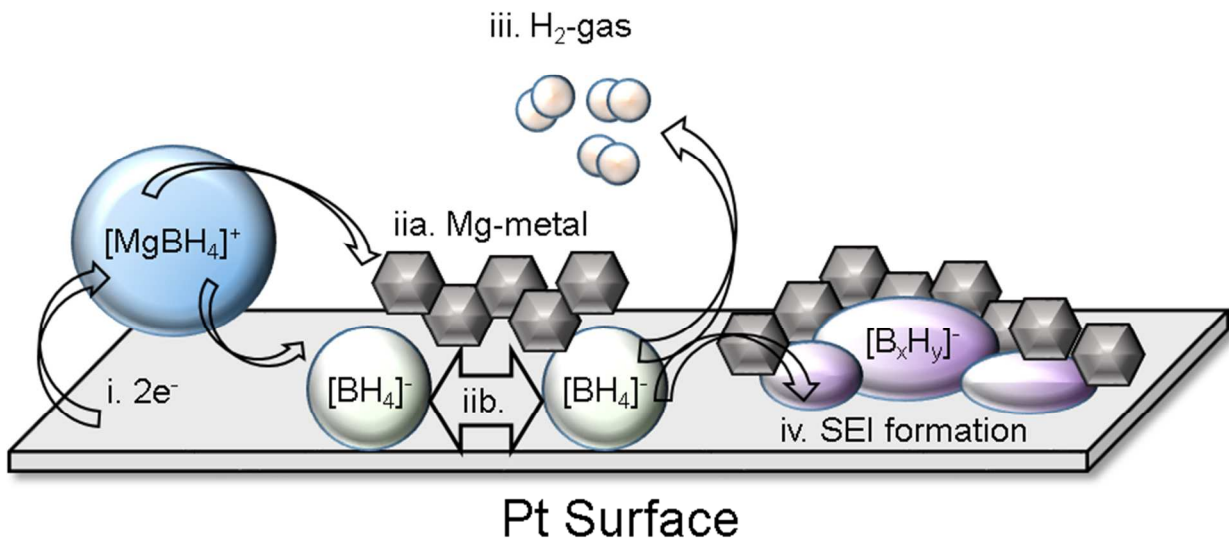


Figure 3. Mg K-edge EXAFS of 1:3 Mg:Li borohydride electrolyte a) As-made and after the 1st-reduction/oxidation cycle. b) *Operando* TEM images of magnesium deposition from 1:3 Mg:Li borohydride electrolyte. c) B K-edge after a 1st cycle and 50th cycle of deposition/dissolution of Mg from 1:3 Mg:Li borohydride electrolyte.

Thus-far, the presence of the $[\text{Mg}(\mu\text{-H})_2\text{BH}_2]^+$ contact-ion pair and weakly coordinated solvent play a large role in the electrochemical reduction of Mg^{2+} , but does not reveal the role of boron as simply a spectator or crucial to the transport properties required of an active SEI layer. Movement of ions through the solid organic interphase requires the presence of an ionically conducting salt, likely containing boron from the BH_4^- anion. Therefore, we turned to B K-edge XAS to determine the role of boron in the interphase layer. Figure 3c) compares the boron K-edge XANES of the reduced and oxidized Pt/ $\text{Mg}(\text{BH}_4)_2$ interface to three reference salts. Due to the lone presence of the σ^* orbital absorption in the $\text{Mg}(\text{BH}_4)_2$ salt, elimination of residual $\text{Mg}(\text{BH}_4)_2$ salt at the interface points to the chemical transformation of BH_4^- anion interphase. One possibility is the oxidation of borohydride to form boron-oxide-type species, i.e. B_2O_3 which shows a clear pre-edge π^* transition at 194 eV, however, referencing to a $\text{Mg}(\text{G}2)(\text{B}_{12}\text{H}_{12})$ [G2 = diglyme] salt shows that interphase layer is structurally similar to hydrogen-based borohydride clusters, and remains at the interphase for 50 cycles. The position of the π^* transition at 190 eV is clear evidence for boron hydrogen clusters formed potentially through **Scheme 1**.



Scheme 1. Interphase formation from $\text{Mg}(\text{BH}_4)_2$ electrolytes.

The key steps are the deposition of an active Mg metal surface (iia) and $[\text{BH}_4]^-$ anion interaction (iib), which promotes the reductive formation of H_2 (iii) gas and higher order boron clusters (iv). The presence of lithium and sodium salts composed of boron cluster anions at the interface of active electrodes has been previously shown to promote the diffusion of Li^+ and Na^+ .²⁶ Therefore, an additional key role for LiBH_4 in the $\text{Mg}(\text{BH}_4)_2$ electrolytes is the increased concentration of BH_4^- anions at the electrified interface promoting steps iib $\rightarrow$ iv. However, the gas-evolution reactions only play a significant role in the initial formation of a SEI and step iv is likely kinetically quenched with continued cycling and larger cluster formation. However, further work, such as porosity measurements, electrical conductivity measurements and spectroscopic measurements must be performed to clarify role of the surface layer. Since the electrochemical oxidative stability of the boron clusters have already been shown to be greater than the BH_4^- anion, future design of the magnesium anode/electrolyte interface should also consider the solid-ionic conductivity of the boron clusters to activate the metal interface.²⁷

Conclusion

As the development of electrochemical storage devices turn to high-energy density metallic anodes, interfacial chemistry and morphology will intimately intertwine to play a pivotal role in the cycle life and performance. For magnesium batteries, the electrochemical capability of borohydride based electrolytes to efficiently deposit and dissolve magnesium metal is challenged by the low oxidative stability of the electrolyte (1.7 V vs Mg). Currently, we are adopting the techniques learned here to gain a deeper understanding of magnesium *monocarborane* in tetraglyme electrolyte (MMC/G4), which displays an oxidative stability is 3.8 V vs Mg without metal corrosion.^{12,13} Here, deep interfacial analysis reveals that magnesium batteries can indeed function with an SEI on the anode surface, where previous research has

shown that magnesium deposition and dissolution can only be achieved without surface films.²

The concert between the active magnesium component, solvent coordination and anion interactions are key to designing future electrolytes and interfaces for rechargeable magnesium batteries.

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Supporting Information

Tabulated EIS fitting elements, figure of *operando* electrochemical/sXAS cell, and magnesium deposition/dissolution cyclic voltammogram of *operando* electrochemical/TEM cell. This material is available free of charge via the Internet at <http://pubs.acs.org>.

AUTHOR INFORMATION

Corresponding Author

*Phone: 734-995-0674. Fax: 734-995-2549. Email: tim.arthur@toyota.com

Notes

The authors declare no competing financial interest

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